

Test practice — the state final attestation format

Thirty questions, forty minutes, no working shown — a different skill from the one practised all year.

LESSON 57–58

2 × 40 MIN

GEOMETRY 11, DTM FORMATI

EXAMINATION TECHNIQUE

LESSON CLOCK

10'

40'

20'

20'

10'

The format	10 min
A practice block of 20	40 min
Marking and analysis	20 min
The four fast techniques	20 min
Homework	10 min

LEARNING OBJECTIVES

- Work a multiple-choice geometry question in under 80 seconds.
- Use elimination, estimation and special cases when a full solution is too slow.
- Recognise the distractors that mark schemes deliberately include.
- Manage the whole paper: order, timing and the last five minutes.

TEXTBOOK

National	pp. 301–310
Cambridge	Core & Extended, pp. 389–392
Workbook	Test paper G1

01

Explanation

The format

FEATURE	WHAT IT MEANS FOR YOU
30 questions, 40 minutes	80 seconds each, on average
four options, one correct	elimination is a legitimate method
no working marked	a correct guess scores the same as a proof
no negative marking	never leave a question blank
calculator usually not allowed	answers are designed to be exact

THE ORDER TO WORK IN

1. **First pass** (20 min): every question you can do in under a minute. Mark the rest.

2. Second pass (15 min): the marked ones, hardest last.

3. Last 5 minutes: transfer everything to the answer sheet, and fill every blank with your best elimination.

A question you cannot do in 80 seconds costs you two you could have done. Leaving it is not giving up; it is arithmetic.

The four fast techniques

1. Elimination by size. A volume must be positive, a probability at most 1, an angle in a triangle under 180° , a surface area larger than any one face. Two of four options usually fail one of these tests.

2. Estimation. $\pi \approx 3$, $\sqrt{2} \approx 1.4$, $\sqrt{3} \approx 1.7$. A cone of radius 5 and height 12 has volume about $\frac{1}{3}(3)(25)(12) = 300$ — enough to pick between 100π and 300π .

3. Special cases. If the answer must hold for every triangle, test the equilateral one. If for every n , test $n = 1$ and $n = 2$. An option that fails a special case is out.

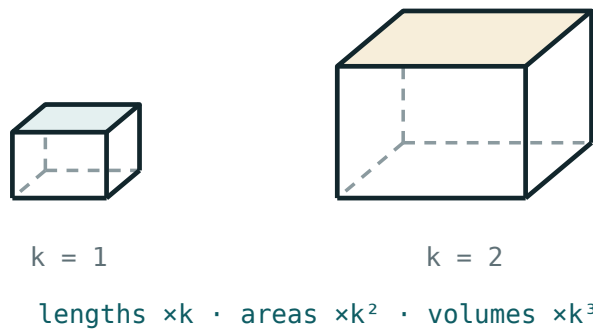
4. Work backwards from the options. Substituting four given answers is often faster than solving. If the options for a radius are 2, 3, 4, 6, computing V for each takes less time than solving a cubic.

QUESTION	FAST ROUTE	TIME
volume of a cone, $r = 5$, $h = 12$	estimate ≈ 314	10 s
surface area of a sphere of radius 3	$4\pi(9) = 36\pi$	5 s
a ratio of similar solids 8 : 27	cube roots 2 : 3	10 s
which solid has $V = \frac{1}{3}Bh$	recall	3 s

The distractors

Wrong options are not random. Each one is the answer to a specific mistake, which means you can recognise your own error in the list.

THE DISTRACTOR IS	THE MISTAKE IT CATCHES
three times the right answer	the missing $\frac{1}{3}$ in a cone or pyramid
half or double	radius used for diameter
the curved area when total was asked	reading
k instead of k^2 or k^3	similar-solids ratios
the slant instead of the height	a cone volume
an answer in cm^3 when litres were asked	units



k, k^2, k^3 — three of the four options in every similarity question.

IF YOUR ANSWER IS EXACTLY ONE OF THE OPTIONS, CHECK ANYWAY

Finding your answer in the list is **not** confirmation — the distractors were built from the likely errors, so a wrong answer is more likely to appear than a random number would be. Spend the last five seconds on the reading check: total or curved? radius or diameter? cm^3 or litres?

A practice block

Twenty questions, twenty-five minutes. Work them in the order given, marking anything that takes longer than a minute.

#	QUESTION	ANSWER
1	Volume of a cube of edge 5	125
2	Space diagonal of that cube	$5\sqrt{3}$
3	Surface area of a sphere of radius 3	36π
4	Volume of a cone, $r = 3$, $h = 4$	12π
5	Slant height of that cone	5
6	Curved area of that cone	15π
7	Volume of a sphere of radius 6	288π
8	Volume of a pyramid, $B = 36$, $h = 5$	60
9	Two similar solids, volumes 8 : 27: length ratio	2 : 3
10	Same pair: area ratio	4 : 9
11	Volume of a cylinder, $r = 2$, $h = 7$	28π
12	Total area of that cylinder	36π
13	Angle of a cube's space diagonal with the base	$\approx 35.3^\circ$
14	Frustum volume, $R = 4$, $r = 2$, $h = 3$	28π
15	Sector angle of the net of a cone, $r = 3$, $l = 9$	120°
16	Capacity in litres of 2000 cm^3	2
17	Curved area of a hemisphere of radius 5	50π
18	Volume of a prism, $B = 15$, $h = 8$	120
19	A cap of height 2 on a sphere of radius 5: its curved area	20π
20	Closed cylinder of least area for a given V : $\frac{h}{r}$	2

MARK IT, THEN COUNT THE TYPES

Do not count the score. Count how many were lost to **reading**, how many to **a formula**, and how many to **arithmetic**. Those three numbers say what to practise; the score says nothing.

02 Worked examples

EXAMPLE 1 A cone has $r = 5$ and $h = 12$. Which is its volume: 100π , 300π , 60π or 900π ?

$$(1) \quad V = \frac{1}{3}\pi(25)(12)$$

$$(2) \quad = 100\pi$$

$$(3) \quad \text{Note the distractor } 300\pi \text{ — the missing } \frac{1}{3}.$$

ANSWER

 100π

EXAMPLE 2 Two similar cones have volumes in the ratio $8 : 27$. What is the ratio of their curved areas?

$$(1) \quad k^3 = \frac{27}{8} \Rightarrow k = \frac{3}{2}$$

Cube root.

$$(2) \quad k^2 = \frac{9}{4}$$

$$(3) \quad 4 : 9$$

The distractors would be $2 : 3$ and $8 : 27$.

ANSWER

 $4 : 9$

EXAMPLE 3 A sphere has surface area 100π . Its volume is: $\frac{500\pi}{3}$, 500π , $\frac{250\pi}{3}$ or 100π ?

① $4\pi R^2 = 100\pi \Rightarrow R = 5$

② $V = \frac{4}{3}\pi(125) = \frac{500\pi}{3}$

③ 500π is the missing $\frac{4}{3}$ distractor.

ANSWER

$$\frac{500\pi}{3}$$

EXAMPLE 4 Without computing exactly, decide whether a cylinder with $r = 7$ and $h = 10$ holds more or less than 1.5 m^3 .

① Units: r and h in cm — read again.

② If in cm: $V \approx 3(49)(10) \approx 1470 \text{ cm}^3$.

③ That is 1.47 litres, not m^3 .

The unit trap, caught by estimation.

ANSWER

About 1.5 litres — the units matter more than the arithmetic

03 Interactive model

Time the class strictly on the twenty questions; the discussion afterwards is about pace, not about geometry.

Twenty questions against the clock

Volume of a cube of edge 5:

Space diagonal of that cube:

Surface area of a sphere, $R = 3$:

Volume of a cone, $r = 3$, $h = 4$:

Curved area of that cone:

Volume of a sphere, $R = 6$:

Similar solids, volumes $8 : 27$: length ratio

Same pair: area ratio

$2 : 3$

$4 : 9$

$8 : 27$

$16 : 81$

Total area of a cylinder, $r = 2$, $h = 7$:

28π

36π

32π

44π

2000 cm^3 in litres:

0.2

2

20

200

Sector angle of a cone's net, $r = 3$, $l = 9$:

90°

120°

180°

60°

Least-area closed cylinder has h r :

1

2

3

$\frac{1}{2}$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Attestation (final test)	Yakuniy attestatsiya	Итоговая аттестация
Multiple choice	Test savoli	Тестовый вопрос
Distractor	Chalg‘ituvchi javob	Отвлекающий ответ
Elimination	Chiqarib tashlash	Исключение
Estimation	Baholash	Оценка
Special case	Xususiy hol	Частный случай
Time per question	Savolga vaqt	Время на вопрос
Answer sheet	Javoblar varaqasi	Бланк ответов

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

With no negative marking you should:

leave hard questions blank

answer every question

guess only at the end

skip the last five

A question taking over 80 seconds should be:

finished anyway

marked and left for the second pass

guessed at once

skipped for ever

A distractor three times the right answer catches:

a unit error

the missing $\frac{1}{3}$

a reading error

nothing

Finding your answer among the options means:

it is right

nothing on its own

check nothing

move on

After marking a practice block, count:

the score

reading, formula and arithmetic errors

the time

the blanks

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Volume of a cube of edge 5

ANSWER 125

2. Space diagonal of that cube

ANSWER $5\sqrt{3}$

3. Surface area of a sphere, $R = 3$

ANSWER 36π

4. Volume of a cone, $r = 3$, $h = 4$

ANSWER 12π

5. Slant height of that cone

ANSWER 5

6. Volume of a prism, $B = 15$, $h = 8$

ANSWER 120

7. 2000 cm^3 in litres

ANSWER 2

Medium · the standard question

7 PROBLEMS

1. Curved area of a cone, $r = 3$, $h = 4$

ANSWER 15π

2. Volume of a sphere, $R = 6$

ANSWER 288π

3. Total area of a cylinder, $r = 2$, $h = 7$

ANSWER 36π

4. Frustum volume, $R = 4$, $r = 2$, $h = 3$

ANSWER 28π

5. Sector angle of a cone net, $r = 3$, $l = 9$

ANSWER 120°

6. Curved area of a hemisphere, $R = 5$

ANSWER 50π

7. Cap of height 2 on a sphere of radius 5: curved area

ANSWER 20π

Hard · stretch and reason

7 PROBLEMS

1. Angle of a cube's space diagonal with the base

ANSWER $\approx 35.3^\circ$

2. Sphere of area 100π : its volume

ANSWER $\frac{500\pi}{3}$

3. Similar solids of volumes $8 : 27$: the area ratio

ANSWER $4 : 9$

4. Least-area closed cylinder: $\frac{h}{r}$

ANSWER 2

5. A cone of radius 5 and height 12 : total surface area

ANSWER 90π

6. A hemisphere and a cone on the same base of radius r have equal volumes: find h

ANSWER $h = 2r$

7. A cube and a sphere have equal surface areas: the ratio of their volumes (cube : sphere)

ANSWER $1 : \sqrt{\frac{6}{\pi}} \approx 1 : 1.38$

07 Homework

Homework — 4 tasks

Time yourself honestly; the timing is the point of the exercise, not the answers.

1. Work the twenty-question block again under timed conditions and record how many took more than 80 seconds.
2. For each question you got wrong, say whether the error was in reading, in the formula, or in the arithmetic.
3. Write out the six distractor types with one example of each from the block.

4. Write your plan for the 40 minutes of the real paper in three sentences: first pass, second pass, last five minutes.